



RAFFLES INSTITUTION

2026 YEAR 6 PRELIMINARY EXAMINATION

Higher 2

MATHEMATICS

9758/01

Paper 1

3 hours

Additional Materials: Printed Answer Booklet
List of Formulae and Results (MF27)

READ THESE INSTRUCTIONS FIRST

Answer **all** questions.

Write your answers on the Printed Answer Booklet. Follow the instructions on the front cover of the answer booklet.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise. Where unsupported answers from a graphing calculator are **not** allowed in a question, you must present the mathematical steps using mathematical notations and not calculator commands.

You must show all necessary working clearly.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **6** printed pages.

RAFFLES INSTITUTION
Mathematics Department

- 1 A curve has parametric equations

$$x = t^2 + 1, \quad y = at^3 + bt^2 + ct,$$

where $t > -2$ and a, b and c are constants.

The curve has stationary points at $t = -1$ and $t = 2$. Given that the curve passes through the point $(5, -2)$, find the values of a, b and c . [4]

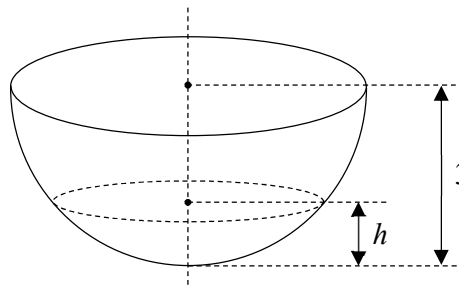
- 2 (a) The sequence $u_1, u_2, u_3, \dots$ is defined by $u_1 = 4$ and $u_{n+1} = -\frac{1}{3}u_n + 2$, for $n \geq 1$.

(i) Write down the value of u_6 . [1]

(ii) Given that as $n \rightarrow \infty$, $u_n \rightarrow l$, find the exact value of l using an algebraic method. [2]

- (b) Using the result $\sum_{r=1}^n r^2 = \frac{n}{6}(n+1)(2n+1)$, find $\sum_{r=0}^n (n+r)(r+1)$, giving your answer in terms of n , in fully factorised form. [3]

3



Water is poured at a rate of 0.5 m^3 per minute into an empty container in the form of an open hemisphere with inside radius 3 m (see diagram). At time t minutes after the start, the radius of the water surface is r m and the volume of the water in the hemispherical container is $\frac{1}{3}\pi h^2(9-h)$, where h m is the depth of the water. When the depth of the water is 1 m,

(a) find the rate of change of the depth of water, [3]

(b) find the rate of change of the radius of the water surface. [3]

- 4 With reference to the origin O , three distinct points A , B and C are such that $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and $\overrightarrow{OC} = \mathbf{c}$. The mid-point of AC is D .

- (a) Express $\overrightarrow{OD}$ in terms of $\mathbf{a}$ and $\mathbf{c}$. [1]
- (b) Given that $|\mathbf{a}| = |\mathbf{c}|$, $|\mathbf{b}| \neq 0$ and BD is perpendicular to AC , show that OB is also perpendicular to AC by using a suitable scalar product. [3]
- (c) Given further that point D lies on line segment OB , $|\mathbf{a}| = 2$, $|\mathbf{b}| = 6$ and $\angle AOC = 60^\circ$, find the exact area of quadrilateral $OABC$. [2]

5 **Do not use a calculator in answering this question.**

It is given that $1 + 2i$ is a complex root of the quartic equation

$$z^4 + pz^3 + qz^2 + rz + 45 = 0,$$

where p , q and r are real numbers.

- (a) State, with justification, another complex root of the equation in the form $a + ib$, where a and b are real numbers and $b \neq 0$. Hence find a quadratic factor corresponding to these two roots. [2]

The quartic equation has a repeated negative real root.

- (b) Find the value of the repeated real root and hence find the values of p , q and r . [4]
- (c) By means of a suitable substitution, solve $45w^4 + rw^3 + qw^2 + pw + 1 = 0$. [2]
- 6 (a) Show that $2x^2 - 4x + 3$ is always positive for all real values of x .
Without using a calculator, solve the inequality $\frac{x^2 - 10x + 3}{(2x + 1)(x - 3)} \geq x$. [6]
- (b) Hence solve the inequality $\frac{x^2 - 10|x| + 3}{(2|x| + 1)(|x| - 3)} > |x|$. [2]

- 7 Conservationists are investigating the effect of habitat loss on the number of deer on an island. They observe that the population of the deer on the island decreases at a rate of 2% every year. The number of deer on the island is P at a time t years after conservationists begin observations.

(a) Write down a differential equation relating P and t . [1]

To maintain population stability, the conservationists relocate deer to the island at a constant uniform rate of h deer per year.

(b) Write down a differential equation to model the new situation. [1]

(c) Without solving the differential equation, determine the value of h if the conservationists would like the number of deer to settle down to 600 after many years. [2]

(d) Solve the differential equation to find an expression for P in terms of t and h . Hence, state the population of the deer, in terms of h , in the long term. [5]

- 8 The curve C has parametric equations

$$x = a \sin t, \quad y = a \sin t \cos t,$$

for $0 \leq t < \pi$ and a is a positive constant.

(a) Sketch the graph of C , stating clearly the coordinates of the points where the curve C meets the axes. [2]

(b) State the equation of the line of symmetry. [1]

(c) Find the equation(s) of the tangent to C which is parallel to the x -axis. [3]

The tangent and normal to C at the point P where $t = \frac{\pi}{6}$ cut the x -axis at points Q and R respectively.

(d) Find the exact area of triangle PQR . [4]

9 A curve C has equation $y = xe^{-x}$, $x \geq 0$.

(a) Show that C has exactly one stationary point. [2]

(b) Sketch C , giving the exact coordinates of the stationary point. [1]

The region R is bounded by C , the y -axis and the line $y = \frac{1}{e}$.

(c) Find the exact area of R . [4]

(d) An ornament is formed by rotating R through 2π radians about the line $y = 2$. By considering a suitable translation of C , find the volume of the ornament. [3]

10 (a) It is given that $f(x) = \frac{x^2 - 20x + 109}{x - 10}$. Describe the transformation which transforms the graph of $y = x + \frac{9}{x}$ onto the graph of $y = f(x)$. [2]

(b) Sketch the graph of $y = f(x)$, labelling clearly the equations of any asymptotes and the coordinates of any axial intercepts and turning points. [3]

(c) On a separate diagram, sketch the graph of $y = f'(x)$, labelling clearly the equations of any asymptotes and the coordinates of any axial intercepts and turning points. [3]

(d) By considering the graph of $(x - 13)^2 + \frac{y^2}{m} = 1$, where $m > 0$, determine the value of m for which the equation $m(x - 13)^2 + \left(\frac{x^2 - 20x + 109}{x - 10}\right)^2 = m$ has exactly one real root. [2]

- 11** With reference to the origin O , the coordinates of A , B and C are $(2, 8, -a)$, $(1, -10, 3)$ and $(5, 6, a)$ respectively. The plane π contains points O , A , B and C .

- (a) Show that $a = 1$. [4]
- (b) Find the acute angle between π and the yz -plane. [2]
- (c) π and the yz -plane meet in the line l . By evaluating an appropriate cross product, find a vector equation for l . [2]
- (d) Write down the distance from A to the yz -plane. Hence or otherwise, find the distance from A to l . [3]

- 12** A seating specialist was tasked to install seats for a performance arena. In the curved section of the arena, he installed 21 seats in the front row facing the arena and for each subsequent row, there are 2 more seats than the row before it. The seating capacity of this curved section is 1664.

- (a) Find the total number of rows in this section of the arena, and the number of seats in the tenth row from the back of the section. [3]

At a performance held at this arena, the cost of a ticket for a front row seat at this curved section is \$151. The cost of a ticket for a seat in every subsequent row decreases by 2% compared to the row before. The cost of tickets for seats in the same row are equal.

- (b) (i) Sam can only afford a ticket below \$100 and he would like to sit as near to the front of the section as possible. Find the row nearest to the front for which he can purchase a ticket. [2]
- (ii) If $F(r)$ is the total cost of tickets for the seats in the r th row from the front of the section, write down $F(r)$ in terms of r . [1]

- (iii) Find the total cost of tickets for all the seats in the first n rows from the front, in the form of

$$143450(1 - A^n) + B \sum_{r=1}^n (rA^{r-1}),$$

where A and B are constants to be determined. [3]

- (iv) Suppose a company can purchase tickets for this performance before public sales are open and they would like to purchase 600 tickets in this curved section with a budget of \$52 000. Determine whether this budget is sufficient for this purchase. [3]